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15 October 1968
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30 October 1953

MILITARY SPECIFICATION

COMPUTER AND TRANSMITTER, TRUE AIRSPEED, PITOT-STATIC, 150-900 KNOTS, TYPE MA-1

1. SCOPE

1.1 This specification covers one type of pitot static true airspeed computer and transmitter, designated Type MA-1.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawing, Engineering and Associated List
MIL-C-5015	Connectors, Electrical, "AN" Type
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-S-6855	Synthetic Rubber Sheets, Strips, Molded or Extruded Shapes
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Specification for
MIL-S-7742	Screw Threads; Standard, Optimum Selected Series, General Specification for
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys

FSC 6610

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-834	Packaging Data Forms, Instructions for Preparation and Use of
MS3102	Connector, Receptacle, Electric, Box Mounting
MS24693	Screw, Machine, Flat Countersunk Head, 100 Deg., Cross Recessed, UNC-2A and UNF-2A
MS33638	Cases, Instruments, Flange-Mounted, Aircraft

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 Qualification. Computers and transmitters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids. (See 4.3 and 6.4)

*3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

*3.3 Materials.

3.3.1 Metals. Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the computer and transmitter, except where magnetic materials are essential.

3.3.3 Protective treatment. When materials are used in the construction of the computer and transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage,

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they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction.

*3.4.1 General design. The computer and transmitter shall conform to MS33638, except as shown on Figure 1. It shall consist of a pitot-static, temperature operated true airspeed computing mechanism operating a synchro control transformer whose electrical characteristics shall be as specified in 3.4.3. The temperature sensing element shall conform to Figure 2 and shall be supplied with the necessary gaskets for flush mounting on the aircraft skin of pressurized cabin aircraft. Unless otherwise specified by the procuring activity, the length of the temperature capillary shall be eight feet. The length of the capillary shall be marked on the case of the computer and transmitter.

*3.4.2 General construction. The computer and transmitter shall be constructed so that no part will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service without failure.

3.4.3 Synchro control transformer. The synchro-control transformer shall be a one phase, three phase control transformer. The input impedance of the three phase winding (line-to-line) shall be 25 ohms or greater at 400 cps. When the three phase winding is excited by 5.9V (line-to-line maximum) the output voltage per degree of error signal shall be at least 0.10V on open circuit. The resistance load across the output required to reduce open circuit output voltage to one half its value shall be 3800 ohms, or less, at 25°C.

3.4.3.1 The synchro shall be connected to the true airspeed mechanism so that the angular rotation of the synchro rotor shall be 36 degrees per 100 knots true. Suitable stops shall be provided to prevent movement of the synchro rotor at speeds below 150 knots or above 900 knots true.

3.4.4 Case. The computer and transmitter case shall consist of a body and cap and shall be in accordance with Figure 1.

3.4.4.1 Body. The body shall be of non-ferrous low density metal, uniform in texture with a smooth surface, and entirely covered with a durable dull black finish.

3.4.4.2 Reinforcement. The inside of the case around each connection shall be of sufficient strength to prevent damage to the case when connections are tightened during installations.

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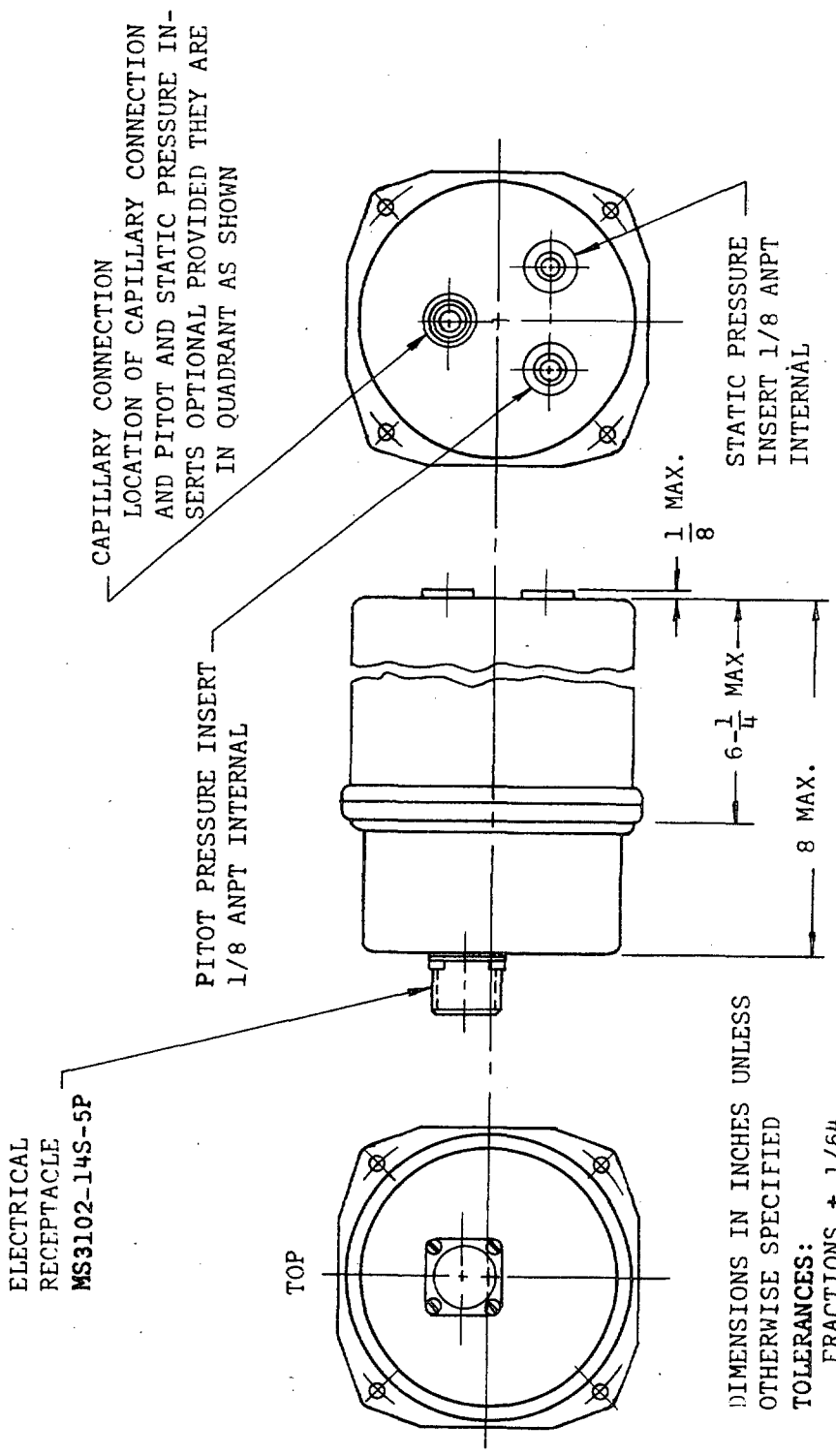


FIGURE 1. Case

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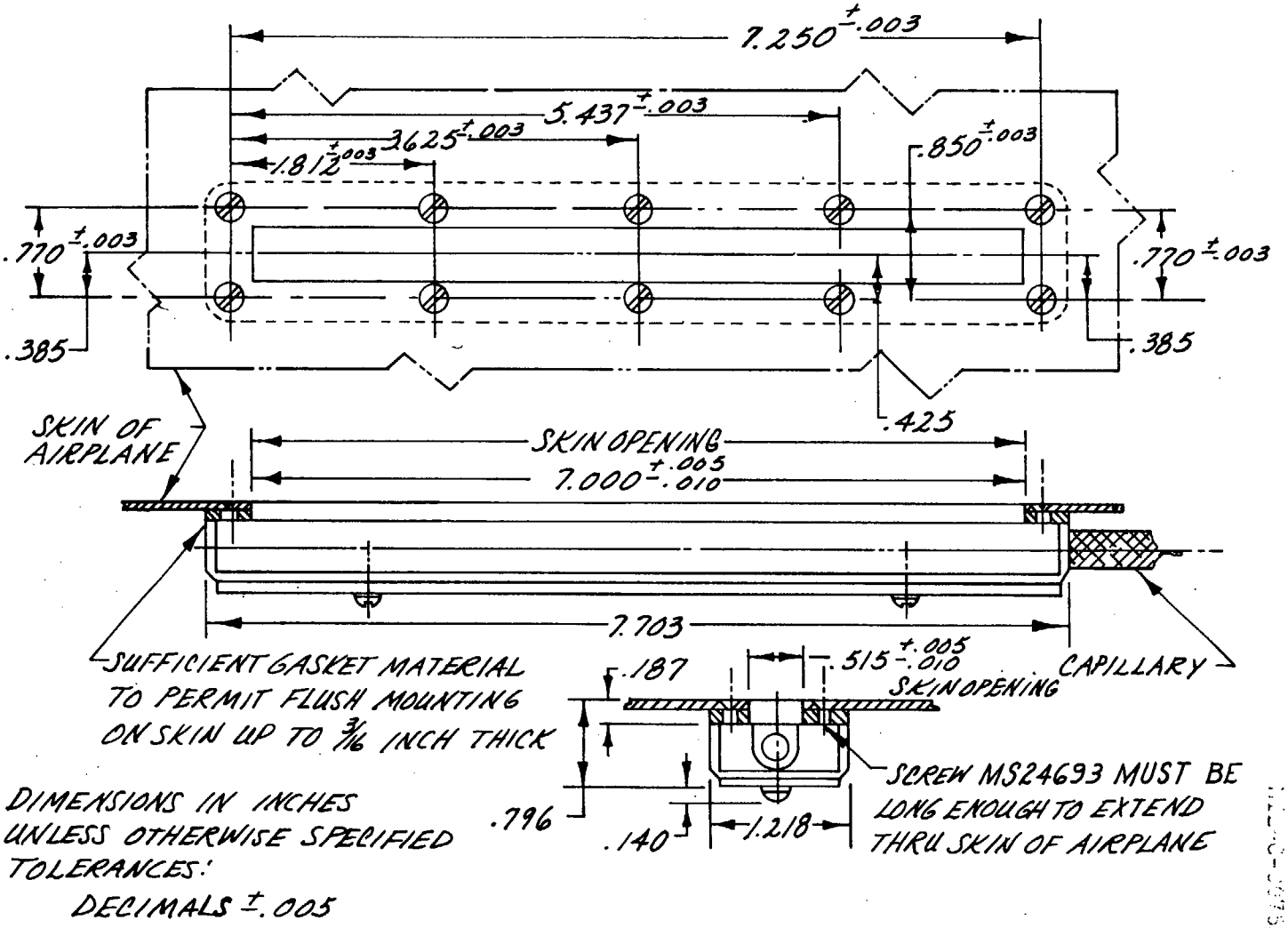


FIGURE 2. Temperature Sensing Element Mounting

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3.4.4.3 Capillary tubing. The free air temperature capillary tubing shall be covered throughout its length with a flexible braided casing. The casing shall be securely attached to the case and to the bulb to protect the capillary tubing against strain.

3.4.4.4 Capillary reinforcement. A suitable flexible reinforcement extending four inches from the case and from the bulb shall be provided on each end of the free air temperature capillary casing. The reinforcement shall be of a flexibility to withstand, without failure, a 180 degree bend of 1-3/8 inches radius. The reinforcement shall be securely attached to the case and to the temperature bulb. The inside diameter of the reinforcement shall be such the reinforcement will slide freely on the capillary casing. The reinforcement shall be braided brass or bronze spiral armor, approximately 9/32 inch outside diameter, plain weave, 7 strands. The brass or bronze wire shall be 0.011 inch in diameter with 6-1/2 to 7 strands per inch.

3.4.4.5 Case marking. Except as specified herein, the exterior of the case shall be finished in durable dull black.

3.4.4.5.1 The back of the case adjacent to the pressure connections shall be marked as follows:

"P" Pressure Connection
"S" Static Connection

3.4.4.5.2 The following shall be marked on the back of the case:

CAUTION
DO NOT BLOW IN TUBES

3.4.5 Indicator. The indicator shall be designed so that proper compensation is made for variation in ambient temperature.

3.4.6 Electrical connector. The electrical connector used in the construction of the computer and transmitter shall be in accordance with MIL-C-5015.

3.4.6.1 Pin connections. The pins of the electrical connector shall be connected to the synchro as follows:

Pin A - Single phase winding No. 1
Pin B - Single phase winding No. 2
Pin C - Three phase winding No. 1
Pin D - Three phase winding No. 2
Pin E - Three phase winding No. 3

The phase rotation shall be C-D-E for airspeed increasing.

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3.4.7 Application of pressure. The pitot pressure shall be applied to the inside of the diaphragm capsule, and the static pressure shall be applied to the interior of the case.

3.4.8 Threads.

3.4.8.1 Screw. Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to MIL-S-7742.

3.4.8.2 Pipe. Pipe threads shall conform to MIL-P-7105.

3.5 Performance. Unless otherwise specified, the computer and transmitter shall be capable of operating satisfactorily under the following conditions:

3.5.1 Temperature. Temperatures ranging from -54 to +71°C.

3.5.2 Humidity. Relative humidity up to 100 percent including conditions wherein condensation takes place in the form of both water and frost.

3.5.3 Salt spray. Exposure to salt sea atmosphere.

3.5.4 Vibration. Vibration incident to service use.

3.5.5 Fungus. Fungus growth as encountered in tropical climates.

3.5.6 Acceleration and shock. Acceleration and shock forces as anticipated.

*3.6 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

3.7 Electrical zero. The electrical zero of the computer and transmitter shall be set at 300 knots true airspeed at an altitude of 15,000 feet and a bulb temperature of 0°C.

3.8 Weight. The weight of the computer and transmitter completely assembled, less mounting nuts and mounting screws, shall not exceed 4 pounds.

3.9 Finishes and protective coatings.

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*3.9.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.9.1.1 Small holes and case inserts need not be anodized.

3.9.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with chemical film in accordance with MIL-C-5541.

3.9.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.9.1.4 Castings containing non-aluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.9.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.9.2 Steel parts. Steel parts shall be cadmium plated, where practicable, in accordance with QQ-P-416, Type II or III, and of a Class that is adequate to achieve the degree of protection required.

3.10 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the computer and transmitter, they shall be held or limited accordingly.

3.11 Installation.

3.11.1 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the screw or bolt or threads.

3.11.2 Mounting screws. The contractor shall furnish sufficient mounting screws for installing the computer and transmitter. They shall be No. 6-32 NC-2A round head brass machine screws having a durable dull black finish. The length shall be sufficient to permit mounting on a panel 3/16 inch thick.

3.11.3 Envelope. An envelope, containing the mounting screws, shall be furnished with each computer and transmitter. The following information shall be printed on the face of the envelope:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS

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3.11.4 Accessories. The contractor shall furnish gaskets with each computer and transmitter and sufficient mounting screws and nuts for installation of the temperature bulb.

3.11.4.1 Gaskets. One gasket constructed from black molded rubber in accordance with MIL-S-6855, Class III, Grade 60, of each type shown on Figure 3 shall be furnished with each computer.

3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.13 Workmanship. The computer and transmitter, including all parts and accessories, shall be finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and freedom of parts from burrs, sharp edges and resin flash that may crumble.

3.14 Cleaning. The computer and transmitter shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign materials after final assembly.

*4. QUALITY ASSURANCE PROVISIONS

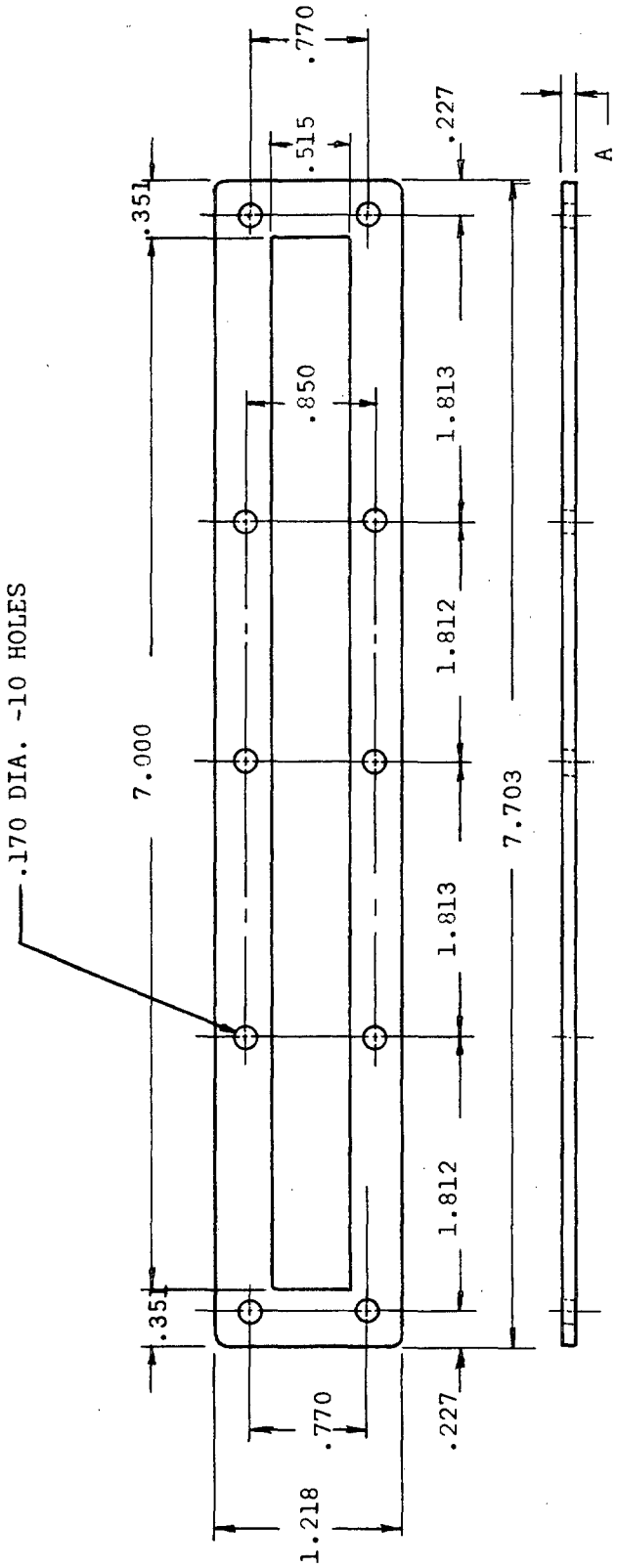
*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of inspection. The inspection of computers and transmitters shall be classified as follows:

a. Qualification inspection (See 4.3)

b. Quality conformance inspection (See 4.4)

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TYPE	A
1	.031 ± .008
2	.062 ± .015
3	.094 ± .015

DIMENSIONS ARE IN INCHES
UNLESS SPECIFIED OTHERWISE.
TOLERANCE:
DECIMALS ± .005

FIGURE 3. Gasket

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*4.3 Qualification inspection. The qualification inspection shall consist of all the inspections and tests of this specification, performed in the following order:

a. Individual tests (See 4.4.1)

b. Sampling tests (See 4.4.2)

c. Environmental tests (See 4.3.2)

*4.3.1 Qualification inspection samples. All of the qualification tests shall be performed on each of five computers and transmitters. These samples shall be identified as qualification samples (See 6.4).

*4.3.2 Environmental tests. The environmental tests shall consist of the tests specified in 4.6.15.1 through 4.6.18.

*4.4 Quality conformance inspection. The quality conformance inspection shall consist of the following:

a. Individual tests (See 4.4.1)

b. Sampling tests (See 4.4.2)

*4.4.1 Individual tests. Each computer and transmitter submitted for acceptance under contract shall be subjected to the following tests:

a. Examination of product (See 4.6.1)

b. Dielectric strength (See 4.6.2)

c. Electrical zero (See 4.6.3)

d. Scale error (See 4.6.4)

e. Friction (See 4.6.5)

f. Leakage (See 4.6.6)

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* 4.4.2 Sampling tests. The sampling tests shall consist of the following:

a.	Weight	(See 4.6.7)
b.	Magnetic property	(See 4.6.8)
c.	Low temperature operation	(See 4.6.9)
d.	High temperature operation	(See 4.6.10)
e.	Seasoning	(See 4.6.11)
f.	Capillary temperature	(See 4.6.12)
g.	Capillary bending and strain	(See 4.6.13)
h.	Vibration error	(See 4.6.14)

*4.4.2.1 Sampling plan. Five computers and transmitters which have been manufactured under essentially the same conditions shall be selected at random from each lot of 500 or fraction thereof on the order and subjected to each of the sampling tests.

*4.4.3 Rejection and retest. When one or more items from a lot fail to pass the sampling tests, acceptance of the lot will be withheld until the extent and cause of failure are determined.

*4.4.4 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the lot will not be made, however, until it is determined that the lot meets all the requirements of the specification.

*4.4.5 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Test conditions.

*4.5.1 Standard atmospheric conditions. Unless otherwise specified, the tests shall be conducted at an atmospheric pressure of approximately 29.92 inches Hg and at a room temperature of approximately 25°C. When tests are performed at atmospheric conditions substantially different from the above values, proper allowance shall be made for the changes in instrument reading.

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4.5.2 Tapping. Unless otherwise specified, the computer and transmitter shall be gently tapped or vibrated before a test reading is taken.

4.5.3 Attitude. Unless otherwise specified, the computer and transmitter shall be tested in the normal operating position.

4.5.4 Vibration. A vibration stand shall be provided which will vibrate at any desired frequency between 5 and 50 cycles per second and shall subject the computer and transmitter under test to vibration such that a point on the case will describe, in a plane inclined 45 degrees to the horizontal plane, a circle, the diameter of which is equal to the amplitude specified herein.

4.5.5 Synchro control transformer connection. Unless otherwise specified, whenever indications from the computer and transmitter are required, the synchro control transformer shall be connected to a follow-up synchro through an amplifier and servo system, as shown in Figure 4. Facilities for determining the rotor position of the follow-up synchro to within ± 0.1 degree shall be provided and the servo system shall not introduce more than 0.1 degree error in the indication. The single phase winding of the follow-up synchro shall be excited with 26V, 400 cps, and the error signal for the amplifier shall be obtained from the single phase winding of the synchro on the computer and transmitter. The indicator shall be set to indicate zero degrees when connected to the computer and transmitter at the electrical zero.

4.6 Test methods.

4.6.1 Examination of product. Each computer and transmitter submitted for acceptance under contract shall be examined to determine conformance with this specification with respect to materials, design and construction, markings, protective coatings and workmanship.

4.6.2 Dielectric strength. A potential of 500V ac at commercial frequency shall be applied between each pin and the case of the computer and transmitter for a minimum period of five seconds. There shall be no breakdown of the insulation.

*4.6.3 Electrical zero. The computer and transmitter shall be tested for electrical zero as follows: Leads from pins C, E, and A shall be tied together. A 0.1 microfarad condenser shall be connected across leads from pins A and B. A voltmeter shall be connected between pins B and D. A voltage of 9V, 400 cycles shall be applied between C and D. With the connections specified in 4.6.4, the bulb temperature shall be adjusted to 0°C and the pressure altitide to 15,000 feet. The differential pressure (indicated airspeed) shall be slowly increased. The voltage should increase as the airspeed

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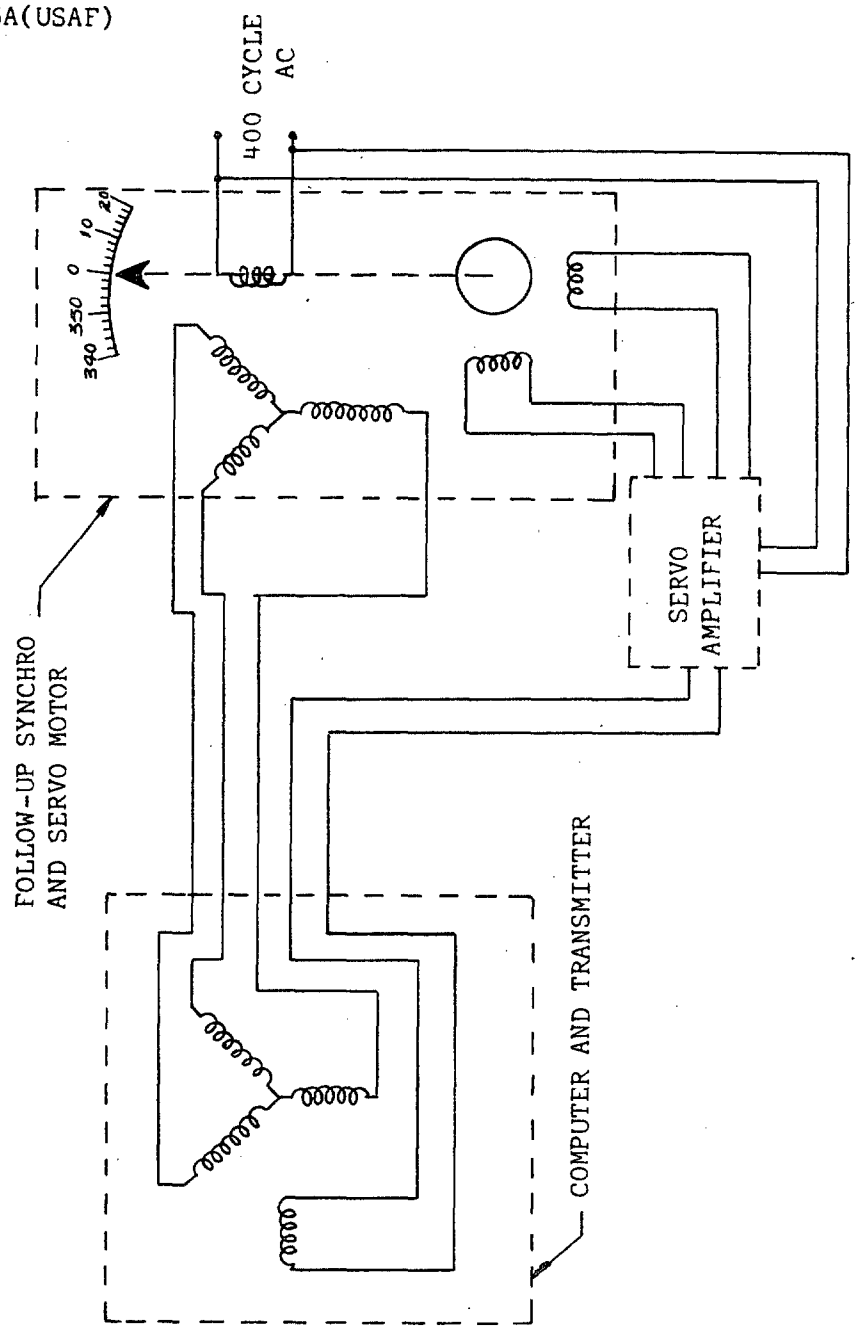


FIGURE 4. Synchro Control Transformer Connection

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is increased from 150 to 200 knots. If the voltage increases, the voltmeter shall be disconnected from pin D and connected to pin A. The indicated airspeed shall then be increased until the voltmeter indicates minimum voltage. This shall be electrical zero. The indicated airspeed at the minimum voltage shall be 238 ± 5 knots.

4.6.4 Scale error. The computer and transmitter shall be mounted on a panel that has facilities for tapping to remove the friction effect. The free air temperature bulb shall be placed in a free air thermometer test fixture in which it is possible to control, vary, and read the temperature accurately throughout the range from -60 to +60°C.

4.6.4.1 The static pressure connection of the computer and transmitter shall be connected by means of a cross to an accurate barometer, a source of air pressure and suction, and to the suction side of a manometer.

4.6.4.2 The pitot pressure connection of the computer and transmitter shall be connected by means of a "tee" to a source of air pressure and suction and to the pressure side of the manometer.

4.6.4.3 The electrical connection shall be as specified in 4.5.5.

4.6.4.4 The computer and transmitter shall be tested for scale error as follows:

4.6.4.4.1 The temperature around the free air temperature bulb in the test fixture shall be adjusted to the first temperature specified in column 1 of Table I.

4.6.4.4.2 The static pressure shall be adjusted to the standard value (29.92 inches Hg) corresponding to zero feet altitude, as specified in column 2 of Table I.

4.6.4.4.3 The pitot pressure shall then be adjusted so that the differential pressure indicated by the manometer is equal to the first test setting, as specified in column 3 of Table I.

4.6.4.4.4 While the temperature of the free air temperature bulb is maintained at the first test temperature setting specified in Table I and the static pressure adjusted to zero feet altitude, the pitot pressure shall be varied so that the differential pressure is equal to each point indicated by an asterisk in column 3. Under these conditions the follow-up synchro indicator shall indicate the angular displacement specified in column 5 and shall not vary from those indications by more than the tolerances listed in column 6. This procedure shall be followed for each of the different pressure altitudes specified in Table I.

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4.6.4.4.5 The procedure specified in 4.6.4.4.4 shall be repeated for the conditions specified in Tables II, III, IV, V, VI, and VII.

TABLE I

Test Temp. °C	Pressure Altitude	Indicated Airspeed Knots	True Airspeed Knots	Indicated Degrees	Tolerance Degrees
+60	0	400*	418	42	4
		450	466	60	4
		500*	514	77	4
		550	560	94	5
		600*	606	110	5
		650	649	126	5
		700*	692	141	6
		750*	733	156	6
		800	773	170	7
		850*	812	184	7
	5,000	450	505	74	4
		500*	553	91	5
		550	601	108	5
		600*	647	125	5
		650*	692	141	6
		700	736	157	6
		750*	779	172	7
		800*	821	187	7
		850	861	202	7
	10,000	500*	594	106	5
		550	644	124	5
		600*	691	141	6
		650	738	158	6
		700*	784	173	7
		750	829	190	7
		800*	872	206	7
	15,000	550	690	140	6
		600*	739	158	6
		650	787	176	7
		700*	835	193	7
		750*	883	220	7

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TABLE I, Cont.

Test Temp. °C	Pressure Altitude	Indicated Airspeed Knots	True Airspeed Knots	Indicated Degrees	Tolerance Degrees
	20,000	600*	790	176	7
		650*	842	195	7
		700*	892	213	7

TABLE II

Test Temp. °C	Pressure Altitude	Indicated Airspeed Knots	True Airspeed Knots	Indicated Degrees	Tolerance Degrees
		150	156	-52	4
		200*	207	-33	3
		250	258	-15	3
		300	308	3	3
		350*	357	21	3
+40	0	400	405	38	3
		450	453	55	4
		500*	498	72	4
		550	544	88	5
		600	588	104	5
		650*	630	119	5
		700	672	134	6
		750	712	148	6
		800*	750	162	7
		850	786	175	7
	5,000	250	281	- 7	3
		300	334	12	3
		350*	387	31	3
		400	438	50	3
		450	488	68	4
		500*	536	85	4
		550	583	102	5
		600	627	118	5
		650*	672	134	6
		700	714	149	6

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TABLE II, Cont.

Test Temp. °C	Pressure Altitude	Indicated Airspeed Knots	True Airspeed Knots	Indicated Degrees	Tolerance Degrees
+40	5,000	750	756	164	7
		800*	796	178	7
		850*	835	192	7

TABLE III

Test Temp. °C	Pressure Altitude	Indicated Airspeed Knots	True Airspeed Knots	Indicated Degrees	Tolerance Degrees
+20	0	150	151	-54	4
		200*	200	-36	3
		250	249	-18	3
		300	298	- 1	3
		350*	345	16	3
		400	392	33	3
		450	438	50	3
		500*	482	65	4
		550	526	81	4
		600	568	96	5
		650*	609	111	5
		700	650	126	6
		750	689	140	6
		800*	726	153	7
		850*	762	166	7
	5,000	150	165	48	4
		200*	219	29	3
		250	272	10	3
		300	324	9	3
		350*	375	27	3
		400	424	45	3
		450	472	62	4
		500*	519	79	4
		550	564	95	5
		600	607	110	5

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TABLE III, Cont.

Test Temp. °C	Pressure Altitude	Indicated Airspeed Knots	True Airspeed Knots	Indicated Degrees	Tolerance Degrees
+20	5,000	650*	650	126	5
		700	691	141	6
		750	731	155	6
		800*	771	169	7
		850*	807	182	7
	10,000	150	181	-43	4
		200*	239	-21	3
		250	297	- 1	3
		300	353	19	3
		350*	407	38	3
		400	459	57	4
		450	509	75	4
		500*	558	93	5
		550	604	109	5
		600	649	126	5
		650*	693	141	6
		700	736	157	6
		750	778	172	7
		800*	818	186	7
		850*	857	200	7

TABLE IV

Test Temp. °C	Pressure Altitude	Indicated Airspeed Knots	True Airspeed Knots	Indicated Degrees	Tolerance Degrees
0	0	200*	193	-38	4
		250	241	-21	3
		300	287	- 5	3
		350*	333	12	3
		400	378	28	3

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TABLE IV, Cont.

Test Temp. °C	Pressure Altitude	Indicated Airspeed Knots	True Airspeed Knots	Indicated Degrees	Tolerance Degrees
0	0	450	422	44	3
		500*	466	60	4
		550	508	75	4
		600	549	90	4
		650*	588	104	5
		700	627	118	5
		750	665	131	6
		800*	701	144	6
		850*	735	156	7
	5,000	150	159	-50	4
		200*	211	-32	3
		250	262	-14	3
		300	312	4	3
		350*	362	22	3
		400	409	39	3
		450	456	56	4
		500*	501	72	4
		550	545	88	4
		600	586	103	5
		650*	627	118	5
		700	667	132	6
		750	706	146	6
		800*	742	159	6
		850*	779	172	7
	10,000	150*	175	-45	4
		200	231	-25	3
		250	287	- 4	3
		300*	340	14	3
		350	393	33	3
		400	443	51	3
		450*	492	69	4
		500	539	86	4
		550	583	102	5
		600*	626	117	5

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TABLE IV, Cont.

Test Temp. °C	Pressure Altitude	Indicated Airspeed Knots	True Airspeed Knots	Indicated Degrees	Tolerance Degrees
0	10,000	650	669	133	6
		700	710	148	6
		750*	751	162	7
		800*	791	177	7
	15,000	150*	192	-39	4
		200	254	-17	3
		250	314	5	3
		300*	371	26	3
		350	371	26	3
		400	480	65	4
		450*	531	83	4
		500	579	100	5
		550	625	117	5
		600*	669	133	6
		650	713	149	6
		700	757	164	7
		750*	799	180	7

TABLE V

Test Temp. °C	Pressure Altitude	Indicated Airspeed Knots	True Airspeed Knots	Indicated Degrees	Tolerance Degrees
-20	0	200*	186	-41	4
		250	232	-25	3
		300	277	- 8	3
		350*	321	8	3
		400	364	23	3
		450	407	38	3
		500*	448	53	3
		550	489	68	4
		600	528	82	4
		650*	566	96	5

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TABLE V, Cont.

Test Temp. °C	Pressure Altitude	Indicated Airspeed Knots	True Airspeed Knots	Indicated Degrees	Tolerance Degrees
-20	0	700	604	109	5
		750	640	122	5
		800*	675	135	6
	10,000	150	168	-47	4
		200*	222	-28	3
		250	276	- 8	3
		300	328	10	3
		350*	378	28	3
		400	427	46	3
		450	473	62	4
		500*	518	78	4
		550	561	94	5
		600	603	109	5
		650*	644	124	5
		700*	684	138	6
	15,000	150	185	-40	4
		200*	244	-20	3
		250	302	7	3
		300	357	20	3
		350*	411	40	3
		400	462	58	4
		450	511	76	4
		500*	557	92	5
		550	602	109	5
		600*	644	124	5
	20,000	150	204	-35	4
		200*	269	-10	3
		250	331	11	3
		300	390	32	3
		350*	447	53	3
		400	500	72	4
		450	551	90	5
		500*	599	108	5
		550*	644	124	5

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TABLE VI

Test Temp. °C	Pressure Altitude	Indicated Airspeed Knots	True Airspeed Knots	Indicated Degrees	Tolerance Degrees
-40	0	200*	179	-43	4
		250	222	-28	3
		300*	265	-11	3
		350	308	3	3
		400*	350	18	3
		450	390	32	3
		500*	430	47	3
		550	469	61	4
		600*	507	74	4
	10,000	150	161	-50	4
		200*	213	-33	3
		250	265	-11	3
		300*	315	5	3
		350	363	23	3
		400*	409	39	3
		450	454	55	4
		500*	498	71	4
	20,000	150	196	-37	4
		200*	258	-57	3
		250	318	6	3
		300*	375	27	3
		350	429	46	3
		400*	480	65	4
		450	529	82	4
	30,000	150	241	-21	4
		200*	315	5	3
		250	384	30	3
		300*	448	53	3
		350	508	75	4
	40,000	150	299	0	4
		200*	386	31	3
		250*	465	59	4
	50,000	150*	369	25	3
		200*	468	60	3

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TABLE VII

Test Temp. °C	Pressure Altitude	Indicated Airspeed Knots	True Airspeed Knots	Indicated Degrees	Tolerance Degrees
-60	0	200*	171	-46	4
		250	212	-32	3
		300*	254	-19	3
		350*	294	- 2	3
	10,000	150	154	-53	4
		200*	204	-35	3
		250*	254	-17	3
		300*	300	0	3
		150	187	-40	4
	20,000	200*	247	-19	3
		250*	304	1	3

4.6.4.4.6 The scale error tests shall be made at every point indicated by an asterisk in Tables I through VII by subjecting the computer and transmitter to the differential pressure specified to produce the readings, first with pressure increasing, then with pressure decreasing. With pressure increasing, the pressure shall be brought up to, but shall not exceed the pressure specified; and with pressure decreasing, the pressure shall be brought down to, but shall not fall below the pressure specified. The errors at the test points shall not exceed the tolerances specified in Tables I through VII.

4.6.5 Friction. The computer and transmitter shall be tested for friction at every point indicated by an asterisk in Tables I through VII. The pressure shall be increased, to bring the pointer approximately to the desired reading, and then held constant while two readings are taken; the first before the computer and transmitter is tapped, and second after the computer and transmitter is tapped. The difference between any two such readings shall not exceed the tolerances specified in Table VIII. The pointer shall move smoothly, while the pressure is varied uniformly, with vibration of the computer and transmitter. This test may be combined with the test for scale errors. During this test, the servo indicator may be tapped to remove any friction in the indicator and servo system.

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TABLE VIII

TESTS	TOLERANCE ±
Friction	2.0 degrees
Position error	2.0 degrees
Vibration error	
Null signal increase	0.2V
Null shift tolerance	2.0 degrees
Vibration failure	2.0 degrees
Scale error	
Low temperature operation	2.0 degrees
High temperature operation	2.0 degrees
Seasoning	2.0 degrees
Overpressure	2.0 degrees
Capillary temperature	1.0 degrees
Capillary bending and strain	1.0 degrees

4.6.6 Leakage.

4.6.6.1 Case leak. The pitot and static pressure connections of the computer and transmitter shall be joined with a "Y" connection and shall be connected together to the same mercury manometer and sources of suction and pressure. A suction of 15 inches Hg shall be applied to the pitot and static pressure connections of the computer and transmitter. With the source disconnected during a period of 10 seconds, the difference between the mercury levels in the manometer shall not change by more than 0.2 inch.

4.6.6.2 Diaphragm capsule leak. The static connection shall be opened to the atmosphere and the pitot connection shall be connected to a mercury manometer and to a source of pressure. A pressure of 20 inches Hg shall slowly be applied to the pitot pressure connection of the computer and transmitter. With the source of pressure sealed during a period of one minute, the difference between the mercury levels in the manometer shall not change by more than 0.1 inch.

4.6.6.3 Position error. The indicator shall be observed while the computer and transmitter is held in any desired position and tapped. This reading shall not differ from the reading when the computer and transmitter is held in any other position, by more than the test amount specified in Table VIII. This test shall be made at every other test point indicated by an asterisk in Tables II, IV, and VII.

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4.6.7 Weight. The computer and transmitter shall be weighed completely assembled, except for connecting fittings, sun shield, and mounting screws. The weight shall not exceed 4 pounds.

4.6.8 Magnetic property. The magnetic property of the computer and transmitter shall be determined in terms of the deflection of a free magnet, approximately 1-1/2 inches long, in a magnetic field with a horizontal intensity of 0.180 ± 0.001 gauss when the computer and transmitter is held in various positions on an east-west line 5 inches from the center of the magnet. An aircraft compass with compensating magnets removed therefrom may be used as the magnet for this test. The maximum deflection of the magnet shall not exceed two degrees. This test shall be conducted with electrical power connected to the synchro.

4.6.9 Low temperature operation. The computer and transmitter shall be placed in a chamber and maintained at a temperature of -54°C for a period of four hours with no electrical power applied. While the computer and transmitter is still at this temperature, power shall be applied; and the computer and transmitter shall meet the requirements specified for scale errors as specified in 4.6.4, except that only Tables VI and VII shall be used. The errors at the test points shall not exceed the tolerances specified in Tables VI and VII by more than the tolerances specified in Table VIII.

4.6.10 High temperature operation. The computer and transmitter shall be placed in a chamber and maintained at a temperature of 71°C for a period of four hours with no electrical power applied. While the computer and transmitter is still at this temperature, power shall be applied; and the computer and transmitter shall meet the requirements specified for scale errors as specified in 4.6.4, except that only Tables II and III shall be used. The errors at the test points shall not exceed the tolerances specified in Tables II and III by more than the tolerances specified in Table VIII.

4.6.11 Seasoning. The pitot pressure connection of the computer and transmitter shall be subjected to 1000 applications of the differential pressure of approximately 20 inches Hg. These applications shall be made at a rate not greater than six per minute. Not less than one hour following this test, the computer and transmitter shall be tested for scale errors as specified in 4.6.4, at 20°C and 5000 feet pressure altitude. The errors at the test points shall not exceed the tolerances specified in Table III by more than the tolerances specified in Table VIII.

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4.6.12 Capillary temperature. A three foot length of capillary shall be subjected to a temperature of 50°C, with the computer and transmitter, the bulb, and the remainder of the capillary at room temperature. The change in reading, due to heating the capillary, shall not exceed the tolerance specified in Table VIII.

4.6.13 Capillary bending and strain. The computer and transmitter indicator reading shall be obtained for a bulb temperature of 50°C. The capillary shall be bent 20 times in one place on a radius of 1/2 inch through an angle of 90 degrees and straightened out after each bending. The flexible braided capillary covering, at points approximately 6 inches from the case and from the bulb, shall be secured in a vise so that the case and the bulb are suspended vertically downward. Care shall be taken to secure the flexible covering lightly in the vise to avoid crushing the covering. The change in reading as the result of bending and straining the capillary shall not exceed the tolerances specified in Table VIII.

4.6.14 Vibration error. The computer and transmitter shall be mounted on a vibration stand with a line joining the centers of the 2 lower mounting holes in a horizontal plane and with the plane of the mounting lugs of the computer and transmitter vertical. This test shall be conducted while differential pressures of 200 and 400 knots are applied to the computer and transmitter. Electrical connections shall be made to the computer and transmitter as shown in Figure 5 and the standard synchro rotated to the null voltage as shown by the lowest amplitude signal on the oscilloscope. The computer and transmitter shall be subjected to vibration with an amplitude between 0.009 to 0.011 inch at frequencies from 500 to 3000 cpm in order to determine that the natural frequency of the computer and transmitter does not occur in this frequency range. While the computer and transmitter is being vibrated, the maximum amplitude of the null signal shall not increase in value more than the amount specified in Table VIII. The angular shift of the standard synchro to maintain the null shall be no greater than the amount specified in Table VIII.

4.6.15 Environmental. The following tests shall be conducted in accordance with the specified procedures of MIL-E-5272, and as specified herein.

4.6.15.1 Low temperature. Procedure II. Before completion of the 24-hour period, pressure shall be applied to the pitot connection and power to the servo system. The computer shall operate without sticking. After the computer has returned to room temperature, it shall meet the requirements specified in 4.6.4. There shall be no damage which would adversely affect subsequent operation.

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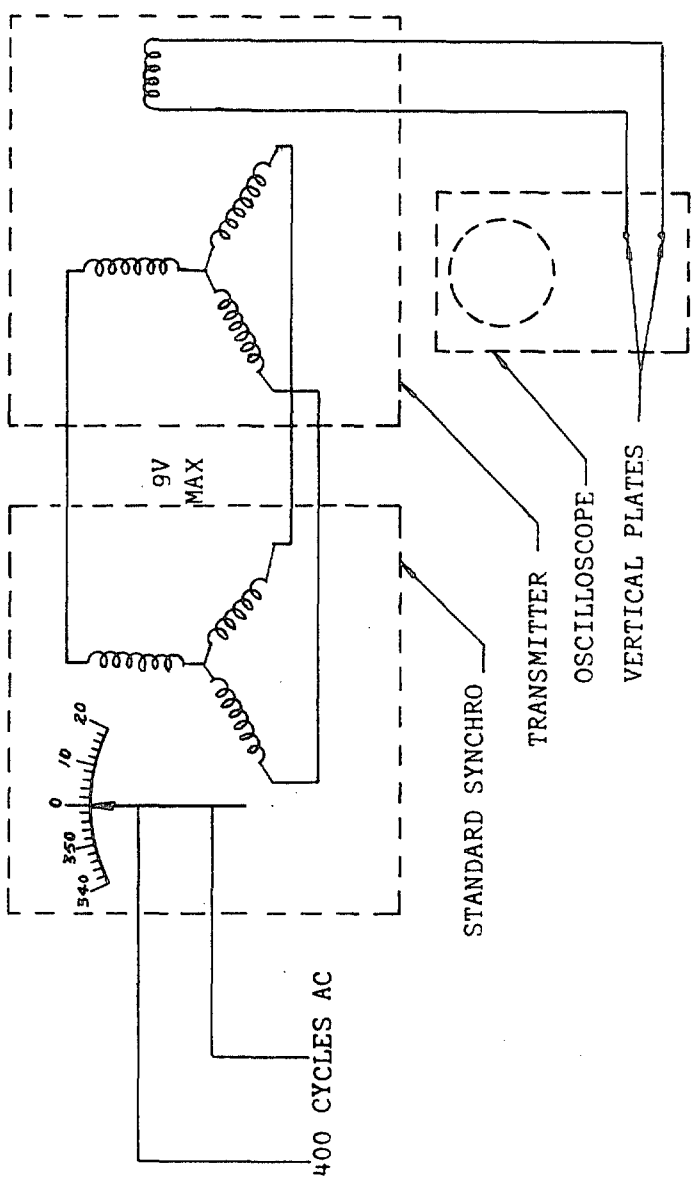


FIGURE 5. Test Setup

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4.6.15.2 High temperature. Procedure I for a period of 24 hours. After return to room temperature, it shall meet the requirements specified in 4.6.4.

4.6.15.3 Vibration. Procedure V. The test shall be conducted while a differential pressure of 400 knots is applied to the computer and transmitter. The difference between the indication of the pointer at the beginning and at the end of the 3-hour vibration period shall not exceed the tolerances specified in Table VIII and no looseness in the mechanism nor damage shall result in any part of the computer and transmitter.

4.6.15.4 Humidity. Procedure I. The computer and transmitter, including applicable external connections shall be mounted in its normal operating position. There shall be no evidence of corrosion or rusting of the internal or external parts of the computer and transmitter.

4.6.15.5 Fungus. Procedure I. The computer and transmitter shall have applicable external connections installed. After the test, the computer and transmitter shall meet the requirements specified for individual tests. There shall be no deterioration which would adversely affect subsequent operation.

4.6.16 Acceleration. The computer and transmitter shall be mounted and accelerated first on its vertical axis and then on each of two axes that are perpendicular to the vertical axis and to each other. The computer and transmitter shall be subjected to an acceleration of 20g in each of the above axes for a period of 30 seconds. At the end of this acceleration, the computer and transmittter shall meet the requirements specified in 4.6.4. No damage to the computer and transmitter shall result from this test.

4.6.17 Salt spray. The salt spray test shall conform to Procedure I of MIL-E-5272 for a period of 50 hours. Upon completion of the 50-hour period, the individual tests shall be repeated. No damage shall result from this test and the tolerances specified for the individual test shall apply.

4.6.18 Overpressure.

4.6.18.1 A pressure of 55 inches Hg shall be applied to the pitot pressure connection of the computer and transmitter for a period of 10 minutes.

4.6.18.2 A pressure of 6 inches of water shall be applied to the static connection of the computer and transmitter for a period of 10 minutes.

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4.6.18.3 Not less than 5 minutes following the application of the overpressure tests, the computer and transmitter shall be tested for scale error as specified in 4.6.4 at 20°C and 5000 feet pressure altitude. The errors at the test points shall not exceed the tolerances specified in Table III by more than the tolerances specified in Table VIII.

4.7 All parts, specimens, or assemblies destroyed in making tests required by this specification, to determine compliance with the specification, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

4.8 Examination of packaging. A random sample of completed packages shall be selected from each lot submitted under contract and examined to insure compliance with section 5 of this specification. Samples shall be selected in accordance with MIL-STD-105, Inspection Level I, AQL 4.0 defective and subjected to applicable tests outlined within the Quality Assurance Provisions of MIL-STD-794.

5. PREPARATION FOR DELIVERY

5.1 General. The preservation, packaging, packing and marking requirements specified herein apply only to direct purchases by, or direct shipments to the government.

5.2 Preservation and packaging. Each computer and transmitter shall be packaged in quantity unit packs of one each in accordance with Level "A" requirements as defined within MIL-STD-794 unless otherwise specified in the contract or order. Where detailed preservation and packaging instructions are not provided in the contract or order, the appropriate preservation method shall be selected in accordance with preservation selection charts contained within MIL-STD-834. Accomplishment of the selected method shall be in accordance with MIL-P-116.

*5.3 Packing. Packing for shipment shall be in accordance with Level "B" requirements of MIL-STD-794, unless otherwise specified in the contract or order.

*5.4 Marking for shipments. In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

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*5.4.1 Reinspection date. Interior packages and shipping containers containing identical items shall bear the legend "REINSPECTION REQUIRED IF IN STORAGE AFTER (Date) ." The date entered shall be determined by adding, to the date of manufacture, the number of months specified in the contract or order for the respective items. Items not contained in a shipping container (unpacked items) shall have firmly fixed thereto an approved tag or label with essentially the same legend.

6. NOTES

6.1 Intended use. The type MA-1 computer and transmitter covered by this specification is intended for use on aircraft to supply signals for indication of true airspeed and for operating various navigation and armament equipment requiring inputs of true airspeed. The true airspeed range is from 150 to 900 knots at altitudes of 0 to 55,000 feet.

6.2 Definitions.

6.2.1 Amplitude. Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.2.2 External connections. When external connections to simulate installed conditions are called for, they shall consist of 10 feet of 1/4 inch copper or aluminum tubing coiled so that moisture can drain out the open end. Similar coils shall be connected to the pitot and static pressure connections.

6.3 True airspeed knots in the tables of this specification are calculated from the following equation:

$$\text{Indication} = .36\ (V_{true} - 300)$$

$$V_{true} = 87.0114\sqrt{T_o\left[\left(1 + \frac{dP}{P}\right)^{0.2857} - 1\right]}$$

Where:

V_{true} = true airspeed in knots

T_o = corrected temperature in degrees K(free air temperature)

$$T_o = 273.2 + T - \left[1.07\ (V_{true})^2 \times 10^{-4}\right]$$

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Where:

T is test temperature in degrees Centigrade
P = pressure corresponding to given altitude
dP = pressure corresponding to given indicated airspeed (same units as P)

*6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the federal government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City Air Materiel Area (OCNE), Tinker AFB, Oklahoma and information pertaining to qualification of products may be obtained from that activity.

NOTE: The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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